

Control of ice thickness in cryo-EM via confinement

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Achieving thin ice with optimal thickness is a prerequisite for successful cryo-EM analysis of embedded macromolecules. Despite decades of effort, precise control of ice thickness in cryo-EM remains a major challenge. Here we reveal that the intrinsic instability of sub-100-nm liquid film, a previously overlooked determinant before vitrification, plays a critical role in controlling ice thickness. We found that well-designed graphene reservoirs can confine liquid film to an optimal thickness and overcome the limitations posed by liquid film instability, achieving precise control of ice thickness with robust reproducibility and large-area uniformity. These graphene reservoirs demonstrate superior liquid retention capabilities than conventional techniques, enabling tunable ice thickness through reservoir depth modulation. Moreover, this approach yields uniform confined ice in graphene sandwiches with controllable thickness and high efficiency, facilitating cryo-EM imaging of a broad range of macromolecule particles with high contrast, reduced motion and improved orientations for more robust high-resolution reconstruction.

Cryo-electron microscopy (cryo-EM) has expanded beyond structural biology into cross-disciplinary domains¹, driving breakthroughs in drug discovery^{2,3}, battery design^{4,5} and materials innovation^{6,7}. In cryo-EM, target samples are embedded in vitreous ice, whose thickness critically dominates the success of cryo-EM analysis because it directly governs the imaging background noise, sample distribution and the attainable resolution of reconstructed structure^{8–10}; however, despite decades of continuous efforts, the control of ice thickness remains unattainable^{11,12}. Current techniques for ice preparation suffer from poor reproducibility

and heterogeneous ice thickness, rendering the thickness control predominantly a trial-and-error process^{13–15}. Given that the vitreous ice is formed by rapid cooling of a thin liquid film, the bottleneck likely lies in the thin liquid film itself, necessitating investigation into the thermodynamic constraints that govern liquid film formation.

Theoretical calculations indicate that a liquid film becomes unstable and undergoes spontaneous thinning once its thickness falls below a critical threshold of ~100 nm^{16,17}, a regime required for high-resolution cryo-EM imaging. This instability arises from liquid migration from

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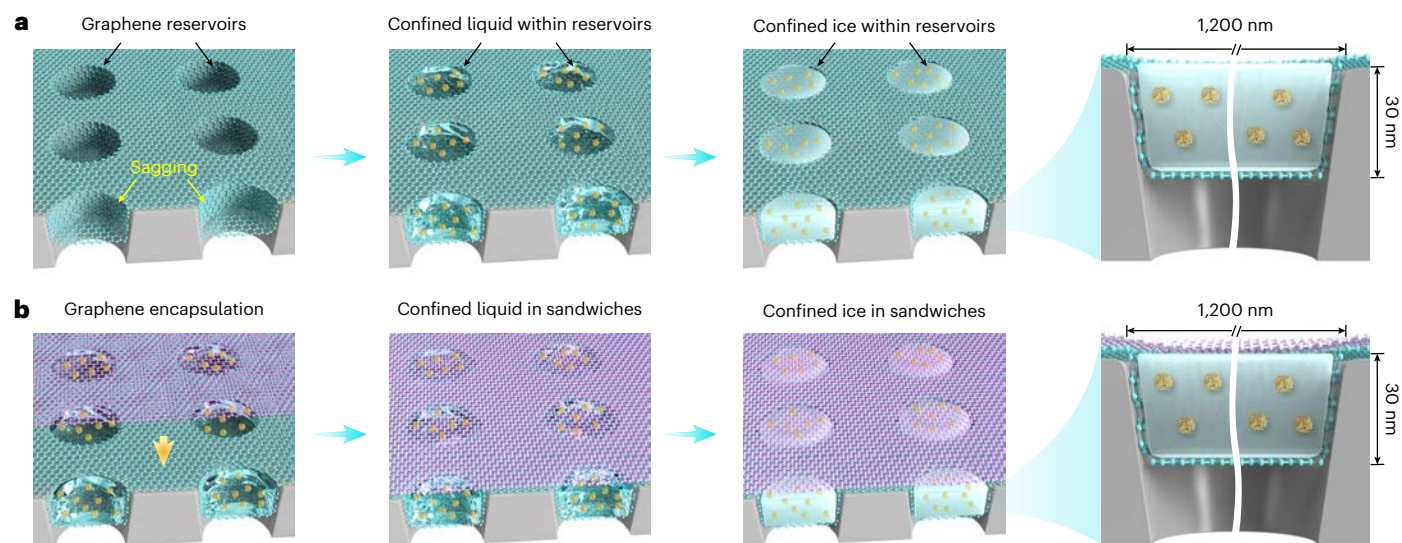


Fig. 1 | Schematic illustration of controlling ice thickness in cryo-EM via confinement. **a**, The adhesion of suspended graphene to the hole sidewall causes the sagging morphology of graphene, forming graphene reservoirs for liquid confinement. After plunge-freezing, uniform vitreous ice layers are prepared with controllable thickness by adjusting the depth of graphene reservoirs. To convey the size correlation between hole diameter and reservoir depth, the right

panel employs a white interval to indicate the representative dimensions. **b**, The liquid is encapsulated by depositing an additional graphene sheet onto the surface of graphene reservoirs, forming confined liquid films in the graphene sandwich structures. Subsequent plunge-freezing yields uniform confined ice with precisely controlled thickness.

thinner regions to adjacent thicker areas, where van der Waals (vdW) interactions are energetically more favorable. As a result, the film becomes increasingly susceptible to rupture and breaks into microscopic droplets in an inherently unpredictable manner¹⁸. The presence of closely packed biological macromolecules has been shown to partially stabilize such thin films^{19–21}, potentially contributing to previously successful structure determinations. Nevertheless, precise control of ice thickness and large-area uniformity remains challenging. These limitations originate from the intrinsic instability of sub-100-nm liquid films, which fundamentally constrains the reliable formation of vitreous ice in cryo-EM.

Recently, liquid under confinement has attracted wide interest due to its emergent physical properties^{22,23}. The confined liquid exhibits significantly slowed molecular migration and a suppressed diffusion coefficient that can be 2–8 orders of magnitude smaller than in bulk^{24–26}. Moreover, the confinement structures can spatially mold the liquid film, even when its thickness is reduced to several atomic layers^{27,28}. These findings suggest that confinement may overcome the thermodynamic limit in preparing sub-100-nm thin liquid film, providing a key to reliable ice formation for high-resolution cryo-EM imaging. The confinement strategy also aligns with the long-pursued goal of sandwiching samples between thin membranes^{12,18}, in which graphene is a promising membrane for liquid encapsulation due to its atomic thickness, excellent flexibility and high conductivity. Unfortunately, the thin liquid film within graphene sandwich encountered the same challenge. Both reproducibility and uniformity are poorly controlled due to the lack of effective control of confinement architectures^{29,30}.

In this work, we have investigated the liquid confinement in reservoirs made of graphene over perforated supporting films, realizing the precise control of ice thickness by adjusting the dimensions and edge profiles of the holes (Fig. 1a). These graphene reservoirs feature a characteristic sagging morphology with controllable depths reaching tens of nanometers. Compared to previously established techniques, the graphene reservoirs demonstrate exceptional liquid retention capabilities, even under extreme specimen preparation conditions. By adjusting the graphene reservoir depth, we can achieve precise control of vitreous ice thickness with both large-area uniformity and

excellent reproducibility. Notably, this approach also enables efficient thickness regulation of confined ice in graphene sandwich configurations (Fig. 1b), overcoming a long-standing challenge that has limited their widespread application in biological and chemical research. The graphene reservoir and derived graphene reservoir sandwich techniques demonstrated high reproducibility in preparing high-quality cryo-EM specimens across diverse macromolecules in our studies, ranging from protein to RNA molecules, enabling high-resolution structural determination.

Results

Design and depth control of graphene reservoirs

Unlike bulk materials, atomically thin graphene over a hole is prone to adhere to the hole sidewalls due to vdW attraction²³, and this adhesion causes graphene sagging (Fig. 1). Previous measurements showed that the sagging depth of graphene over holes (diameter ~1 μm) ranged from 2 nm to 30 nm^{31–33}, primarily determined by the uneven hole edge morphology and vdW interaction strength³⁴. To achieve uniform sagging depth, we fabricated identical holes with smooth, rounded edges (Fig. 2a, Extended Data Fig. 1, Supplementary Figs. 1 and 2 and Supplementary Video 1). Graphene membranes transferred onto these holes sag inward, forming structures referred to as graphene reservoirs (Fig. 2b and Supplementary Fig. 3). Atomic force microscopy (AFM) revealed that graphene membranes of graphene reservoirs were stretched tautly across the hole edges, and the sagging depth was 30 ± 1 nm in 1,200-nm diameter holes (Fig. 2b). Statistical analysis further confirmed the uniformity of sagging depth (Fig. 2c), demonstrating the effectiveness of our design on hole morphology for reliable graphene reservoir formation. In contrast, commonly used EM grids (Quantifoil) feature sharply protruding hole edges (Fig. 2d)³⁵, causing graphene membranes to bulge outward over the holes after transfer (Fig. 2e). The bulging height reaches up to 44 ± 5 nm (Fig. 2f), which poses challenges for controlling ice thickness, as discussed subsequently.

Notably, the dimensions of hole arrays can be precisely controlled. The overall array size has been scaled up to 4 inches, enabling batch production of over 500 homemade grids (Fig. 2g). By varying the hole diameter from 600 nm to 2,000 nm, we achieved controllable graphene

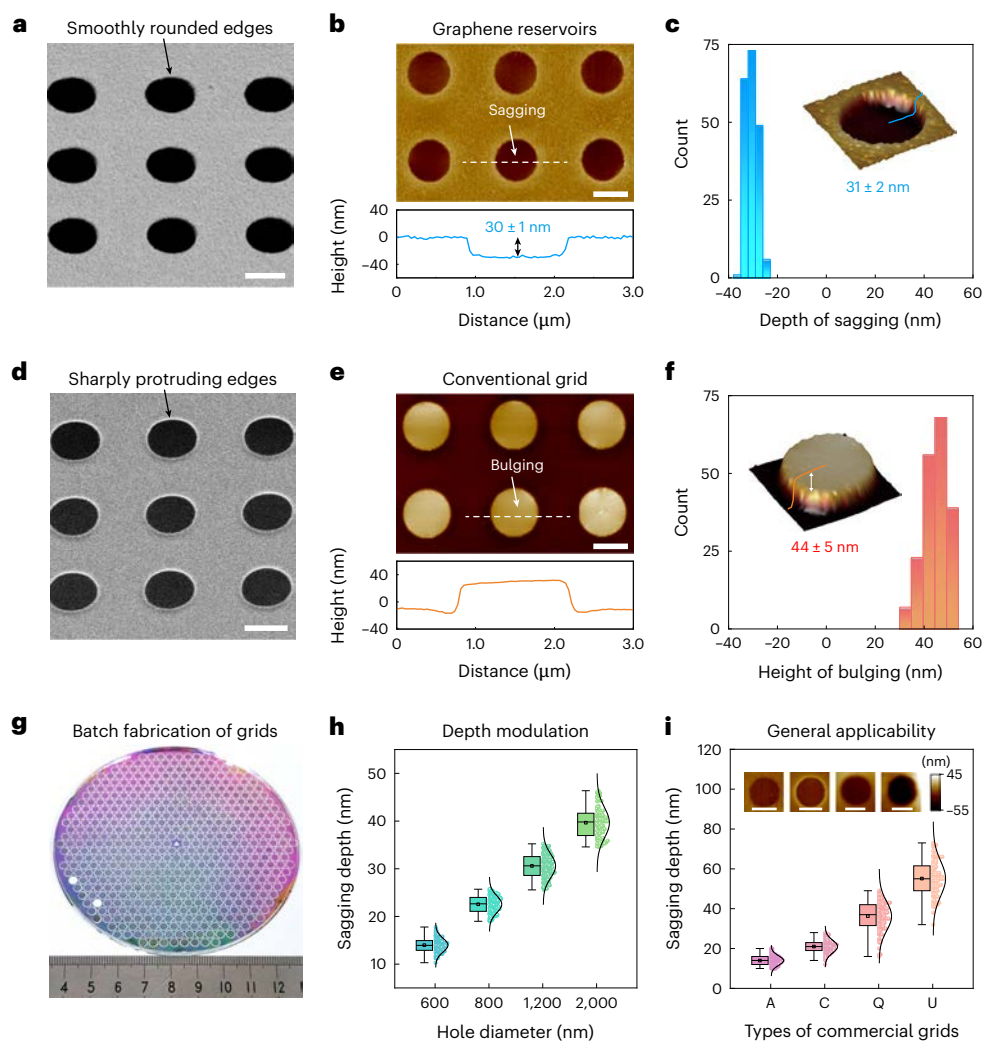


Fig. 2 | Design and depth control of graphene reservoirs. **a**, Tilted SEM image revealing the smoothly round edges of homemade grid. **b**, Representative AFM image reveals that graphene membranes over the holes exhibit a sagging morphology, with a depth of around 30 nm measured from the depth profile, forming graphene reservoirs. **c**, Sagging depth distribution of graphene reservoirs (31 ± 2 nm). Inset shows 3D sagging morphology of graphene reservoir characterized by AFM. **d**, Tilted SEM image revealing the sharply protruding edges of conventional grids (Quantifoil). **e**, Typical AFM image shows that graphene membranes over conventional grid holes exhibit a bulging morphology. **f**, Bulging height distribution of conventional grids (44 ± 5 nm). Inset shows 3D bulging morphology of conventional grid characterized by AFM. **g**, Optical image showing the batch production of over 500 homemade

grids. **h**, Tunable depths of homemade graphene reservoirs achieved by varying the hole diameter: 600 nm ($n = 150$), 800 nm ($n = 179$), 1,200 nm ($n = 193$) and 2,000 nm ($n = 100$). **i**, Sagging depths of graphene reservoirs fabricated with commercially available grids, including ANTcryo grid (A, $n = 131$), C-flat grid (C, $n = 111$), backside of Quantifoil grid (Q, $n = 100$) and backside of UltrAufoil grid (U, $n = 100$). Insets show the corresponding AFM images, revealing the sagging morphologies of graphene on each grid type. For the box-and-raincloud plots in (**h**) and (**i**), the box limits indicate the first and third quartiles (25th and 75th percentiles), the center line indicates the median, the black square indicates the mean, the whiskers indicate the minimum and maximum values, individual data points are shown as jittered points, and the half-violin represents the kernel density estimate. Scale bars, 1 μ m.

sagging depths of 14 ± 1 nm, 23 ± 2 nm, 31 ± 2 nm and 40 ± 3 nm, respectively (Fig. 2h and Extended Data Fig. 2). These experiment results were supported by our theoretical model (Supplementary Fig. 4), in which sagging depth w_0 can be described by

$$w_0 = kD^{\frac{2}{5}}R_c^{\frac{3}{5}}\Gamma^{\frac{2}{5}}E_{2D}^{-\frac{2}{5}} \quad (1)$$

where k is a constant, D is the hole diameter, R_c is the curvature radius of the hole edge, Γ represents the adhesion energy between graphene and hole edge, and E_{2D} is the Young's modulus of graphene, respectively. Therefore, increasing D and enlarging R_c (which corresponds to smoother edge morphology) contribute to enhanced sagging depth w_0 .

With these insights, we further investigated the compatibility of the graphene reservoir strategy across various commercially available

grids, including ANTcryo grid, C-flat grid, backside of Quantifoil grid and backside of UltrAufoil grid (Fig. 2i and Supplementary Video 2). All these grids displayed smooth hole edges under scanning electron microscopy (SEM) characterization (Extended Data Fig. 3). After transferring graphene onto these grid sides, we observed similar graphene sagging phenomena. Specifically, the graphene sagging depths of ANTcryo grid, C-flat grid, backside of Quantifoil grid and backside of UltrAufoil grid are 14 ± 2 nm, 21 ± 3 nm, 36 ± 7 nm and 55 ± 9 nm, respectively (Fig. 2i). These results demonstrate that our confinement strategy enabled by controlled graphene reservoirs, can be broadly applicable in cryo-EM labs using commercially available grids.

Precise control of ice thickness in cryo-EM

The sagging morphology of graphene reservoirs, naturally creating localized containers, inspired us to investigate their liquid preservation

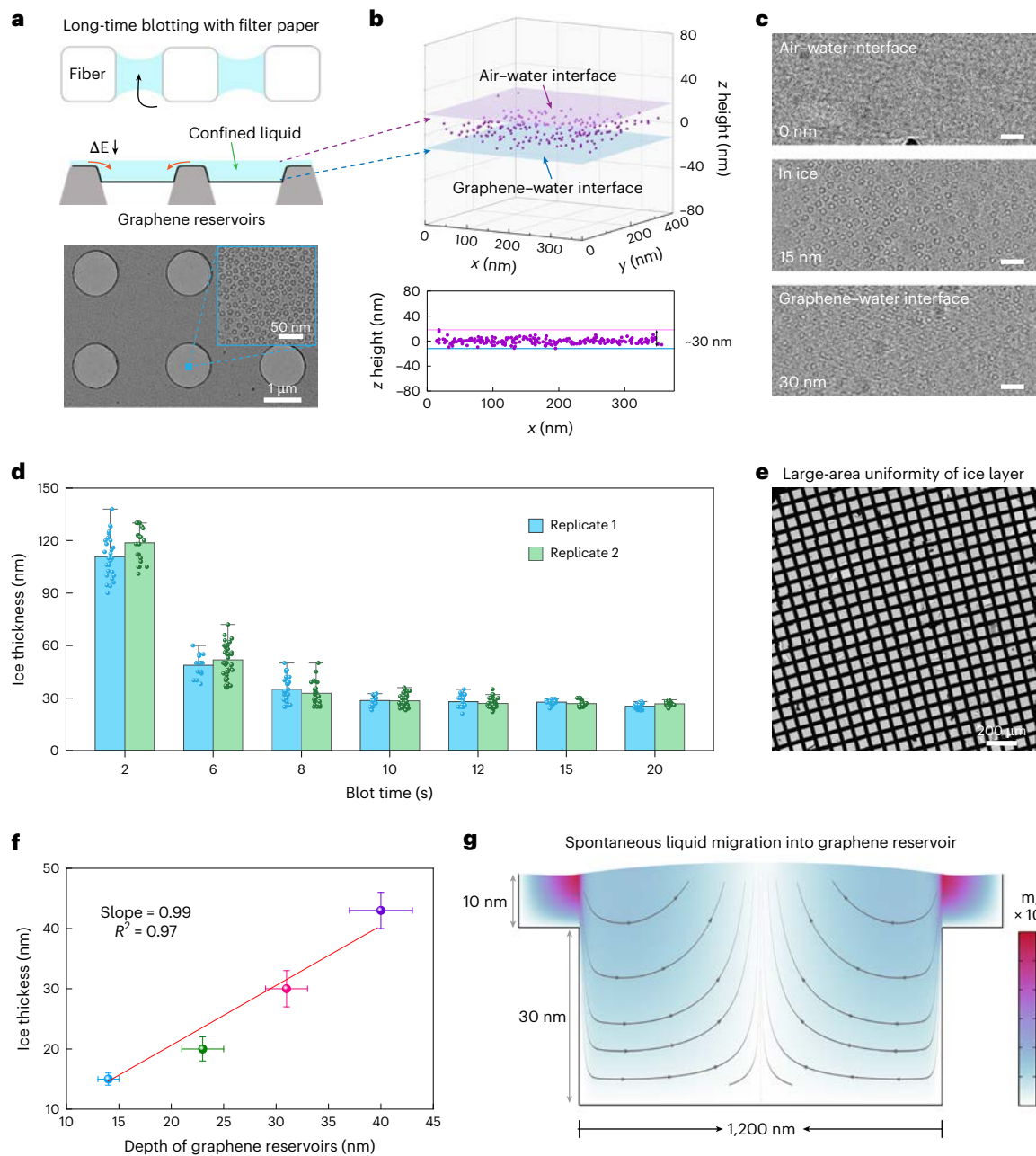


Fig. 3 | Precise control of ice thickness using graphene reservoirs. **a**, The schematic shows that after long-time blotting with filter paper (top), a confined liquid film is generated in the graphene reservoir (middle). Typical cryo-EM image showing apoferritin particles embedded in the confined vitreous ice (bottom). **b**, Spatial distribution of apoferritin particles within the confined ice layer. 3D perspective (top); 2D perspective (bottom). Each purple spot represents one individual protein particle. **c**, Typical cryo-ET slices extracted from the cryo-ET reconstruction of the cryo-specimen showing the air–water interface (top), in-ice layer (middle) and graphene–water interface (bottom), respectively. Scale bars, 50 nm. **d**, Statistical analysis of ice thickness generated with different blot times, demonstrating reliable reproducibility. At longer blot times (8–20 s), the ice thickness corresponds with the depth of the graphene reservoirs. Average ice thicknesses from two independent experimental replicates are comparable as

follows: 2 s, 111 ± 12 nm ($n = 29$) and 119 ± 9 nm ($n = 24$); 6 s, 49 ± 6 nm ($n = 15$) and 52 ± 9 nm ($n = 40$); 8 s, 35 ± 7 nm ($n = 30$) and 33 ± 6 nm ($n = 24$); 10 s, 29 ± 3 nm ($n = 19$) and 28 ± 3 nm ($n = 43$); 12 s, 28 ± 4 nm ($n = 19$) and 27 ± 2 nm ($n = 45$); 15 s, 28 ± 1 nm ($n = 26$) and 27 ± 2 nm ($n = 24$); and 20 s, 25 ± 1 nm ($n = 27$) and 27 ± 1 nm ($n = 22$). **e**, Typical overview atlas of the cryo-specimen showing the large-area uniformity of ice layer. **f**, Tunable ice thickness by controlling the depth of graphene reservoirs. Average ice thicknesses closely match the corresponding graphene reservoir depths. Ice thicknesses are 15 ± 1 nm ($n = 16$), 20 ± 2 nm ($n = 59$), 30 ± 3 nm ($n = 29$) and 43 ± 3 nm ($n = 46$), whereas the corresponding graphene reservoir depths are 14 ± 1 nm ($n = 150$), 23 ± 2 nm ($n = 179$), 31 ± 2 nm ($n = 193$) and 40 ± 3 nm ($n = 100$). **g**, Numerical simulation reveals spontaneous liquid migration into graphene reservoir from thin peripheral regions. The colors denote differences in flow velocity and the arrows show the flow direction.

capabilities. To test this, we employed the standard blotting method to our graphene reservoir-equipped grids (1,200-nm diameter holes) for ice layer preparation. As shown in Fig. 3a, excess liquid was wicked away from graphene reservoir surface using filter paper to form a confined liquid film for subsequent ice formation. Cryo-electron tomography (cryo-ET) revealed that this confined liquid film indeed

exhibited a sagging morphology within the graphene reservoir (Supplementary Fig. 5). Of note, we found that the confined liquid film remained stable even after a significantly extended blot time of 10 s, which is 5–10 times longer than typical blotting durations in our usual practice (usually 1–2 s) on conventional graphene grid. Protein particles were well-dispersed in such vitrified liquid film and preserved their

native structures (Fig. 3a). The proteins' three-dimensional (3D) spatial distributions within the vitreous ice were resolved using single-particle analysis (Fig. 3b) and cryo-ET (Fig. 3c). As showed by the results, most protein particles were adsorbed at the graphene–water interface (Fig. 3b,c), away from the detrimental air–water interface, which is known to cause protein denaturation and preferential orientation^{12,36,37}. Both single-particle analysis and cryo-ET measurements revealed a consistent ice thickness of ~30 nm (Fig. 3b,c and Supplementary Fig. 6), closely matching the sagging depth of the graphene reservoir (Fig. 2c), revealing the graphene reservoir's capability for ice thickness control.

We further evaluated the reproducibility and large-area uniformity of ice thickness prepared with graphene reservoirs. We used cryo-ET reconstruction to measure the ice thickness across different regions of multiple specimens prepared with varying blot times. The average ice thickness was 111 ± 12 nm, 49 ± 6 nm, 35 ± 7 nm, 29 ± 3 nm, 28 ± 4 nm, 28 ± 1 nm and 25 ± 1 nm for blot time of 2 s, 6 s, 8 s, 10 s, 12 s, 15 s and 20 s, respectively (blue data; Fig. 3d and Supplementary Fig. 7). Independent replicate experiments under identical conditions produced comparable results (119 ± 9 nm, 52 ± 9 nm, 33 ± 6 nm, 28 ± 3 nm, 27 ± 2 nm, 27 ± 2 nm and 27 ± 1 nm; green data; Fig. 3d and Supplementary Fig. 8), demonstrating the reliability of graphene reservoirs in controlling ice thickness. Notably, the ice thickness exhibited no significant decrease beyond 10 s of blotting, maintaining a value of approximately 27 ± 3 nm. Similar effects were also observed when using grids with hole diameters of 600 nm and 2,000 nm, with the former stabilizing at a thinner ice layer and the latter at a thicker layer after long-duration blotting (Extended Data Fig. 4). This means the graphene reservoir design provides a robust platform for reproducible thin ice preparation. Besides, the confined ice layer exhibited uniform ice thickness across the entire grid, as revealed by the overview atlas of the grid with homogeneous brightness and unnoticeable variability (Fig. 3e).

We then explored if the plateau thickness of ice can be tailored by adjusting the sagging depths of graphene reservoirs. Indeed, controllable ice thicknesses of 15 ± 1 nm, 20 ± 2 nm, 30 ± 3 nm and 42 ± 3 nm were achieved using graphene reservoirs with hole diameters ranging from 600 nm to 2,000 nm (Fig. 3f and Extended Data Fig. 5). The measured ice thickness closely matched the sagging depths of their respective graphene reservoirs (Fig. 2h), further confirming our confinement approach for reliable ice thickness control. The controlled thin ice helps improve image quality, as evidenced by the reduced B factor and decreased particle motion (Supplementary Fig. 9). In contrast, the conventional grid with protruding holes suffered from rupture of the liquid film under the same blot time of 10 s (Supplementary Fig. 10a,b). The ruptured liquid film and resultant uneven ice caused the protein aggregation and denaturation, leading to a significant reduction of usable particle percentage for cryo-EM reconstruction (Supplementary Fig. 10c,d). These differences originate from the thermodynamic behavior of a sub-100-nm liquid film. The free energy of such film exhibits positive correlation with the long-range contribution of vdW force (P), as described by¹⁶:

$$P(h) = \frac{A}{12\pi h^2} \quad (2)$$

where h is the thickness of liquid film and $A > 0$ for a wetting liquid. According to this equation, thinner films experience a higher vdW force, prompting spontaneous liquid migration toward thicker areas to achieve energy minimization. This mechanism also explains how graphene reservoirs retain liquid under extreme blotting conditions: liquid spontaneously migrates from thin peripheral regions into the thicker graphene reservoir domains (Fig. 3a). To make it more visible, we performed theoretical simulations to reveal the diffusion behavior of liquid film around the graphene reservoirs. As shown in Fig. 3g, liquid flow velocity in thin peripheral regions significantly exceeds that in thicker graphene reservoir areas, directly driving liquid influx into graphene

reservoirs. This phenomenon consistently appears across graphene reservoirs with different sagging depths (Supplementary Fig. 11).

Ice confinement in graphene reservoir sandwich

The successful ice thickness control of graphene reservoirs provides new opportunities for a more challenging case—achieving the precise control of ice thickness confined in graphene sandwich structures (Fig. 4a). The graphene sandwich technique has garnered significant attention across materials science, chemistry and biology over the past decade, owing to its great potential to resolve the native structures and dynamics of single atoms, biomolecules and nanoparticles^{38,39}; however, its widespread adoption has been limited by long-standing challenges in controlling specimen thickness and uniformity^{11,24,29}. In light of this, graphene reservoirs can provide three key advantages: uniform dimensions, periodic arrangements and thermodynamically favorable liquid migration, collectively making the formation of sandwiched liquid more controllable. As expected, the obtained confined ice in graphene reservoir sandwich is uniform (Fig. 4a), exhibiting a more homogeneous electron transmission than that in conventional graphene sandwich (Fig. 4b). Moreover, the large-area uniformity of confined ice can be well reproduced, as confirmed by the overview atlases of multiple grids (Fig. 4c and Supplementary Fig. 12). The robust reproducibility benefits from graphene reservoirs' excellent liquid retention capability during extended blot time. Detailed step-by-step procedures have been provided (Methods) to enable experimental replication. Most procedures are compatible with commonly used blotting method in cryo-EM specimen preparation, facilitating broad and immediate applicability.

We further analyzed the distribution of protein molecules within the confined ice by the graphene reservoir sandwich and measured the ice thickness using cryo-ET. As shown in Fig. 4d,e, the 20S proteasome particles predominantly adsorbed at both top and bottom graphene surfaces. The measured ice thickness was ~30 nm, which was suitable for embedding proteins without introducing extra background signal. The thickness statistics further demonstrated the uniformity of confined ice, which exhibited a narrow distribution centered at 30 nm and matched the sagging depth of graphene reservoirs (Fig. 4f and Supplementary Fig. 13). In contrast, conventional graphene sandwiched ice showed larger thickness (52 ± 18 nm) with a high degree of variability (Fig. 4f and Supplementary Fig. 14). Such thickness variation represents a major limitation for the broad application of the conventional graphene sandwich technique, as excessively thin ice layers compromise protein structural integrity, while overly thick ice layers obscure protein particles in cryo-EM imaging. Moreover, we found that the thickness of confined ice in graphene reservoir sandwich could also be precisely controlled by varying the layer number of graphene reservoir. Increasing from a monolayer to a trilayer causes a threefold increase in E_{2D} (Eq. (1)), which inversely reduces the ice thickness from 30 ± 1 nm to 19 ± 2 nm (Supplementary Fig. 15). This thickness reduction by $3^{-2/5}$ times aligns closely with model predictions.

Beyond ice thickness control, graphene reservoirs can enhance the efficiency in fabricating confined ice in sandwich. Quantitative analysis revealed a success rate of over 90% across 2,219 grid squares using our graphene reservoirs versus ~67% for conventional method ($n = 1,973$ squares; Fig. 4g and Supplementary Fig. 16). This high efficiency (>90%) ensures successful encapsulation of macromolecules, minimizes costly retrials and enables high-throughput cryo-EM data collection. We further compared the beam-induced motion between different kinds of grids. The particle motion is mainly caused by beam-induced ice bulking during imaging⁴⁰, which degrades image quality and impairs high-resolution information in cryo-EM. Particles in graphene reservoir sandwich exhibited reduced motion amplitude and enhanced convergence compared to those in conventional sandwiched ice (Fig. 4h). Similar phenomena were observed when comparing the particle motion between graphene reservoir and conventional graphene without sandwiching (Supplementary Fig. 17). Moreover,

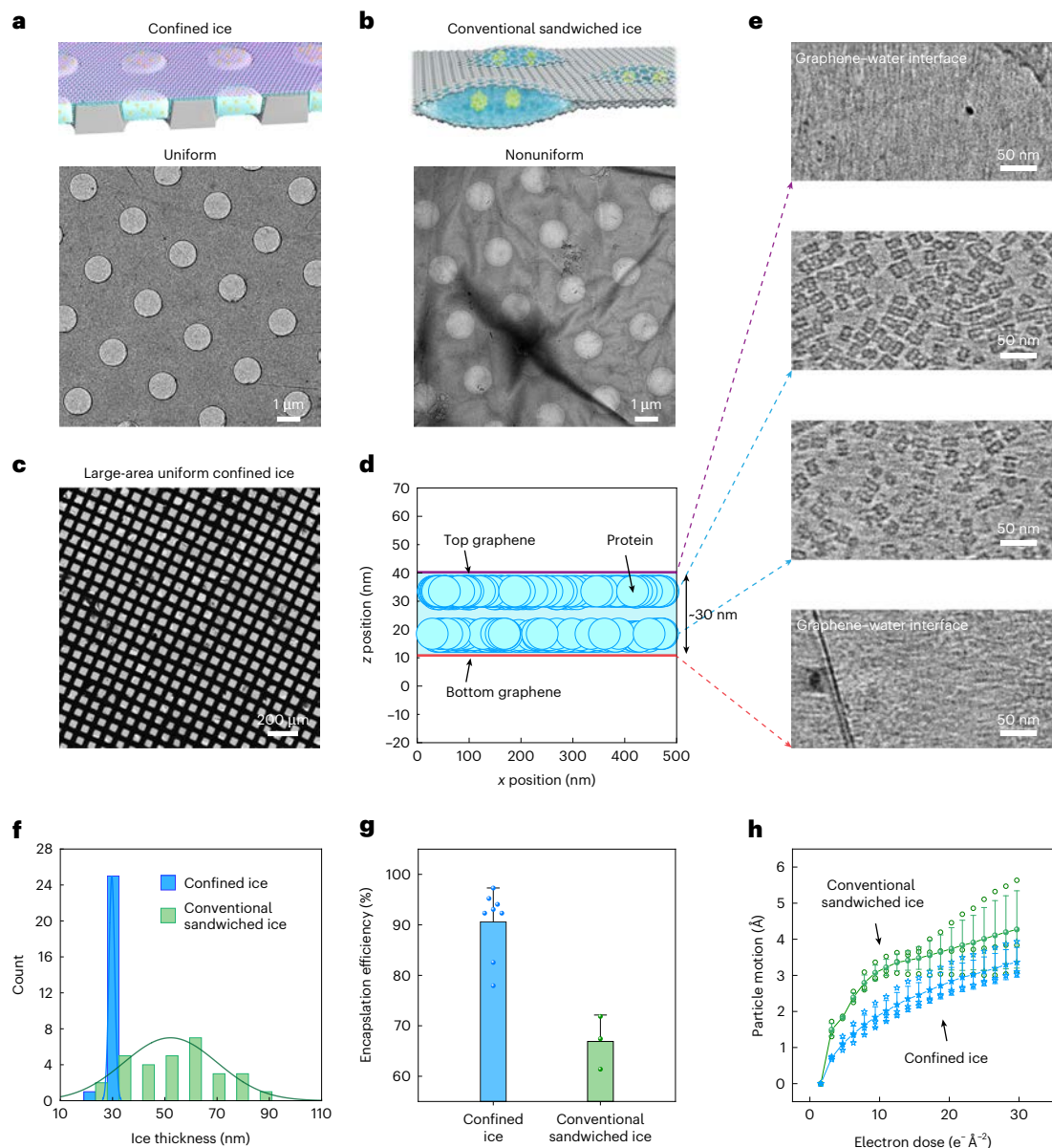


Fig. 4 | Uniform confined ice in graphene reservoir sandwich with controllable thickness. **a,b**, Schematic illustrations and corresponding cryo-EM images showing uniform confined ice in graphene reservoir sandwich (**a**), and heterogeneous ice thickness for conventional sandwiched ice (**b**), respectively. **c**, Typical overview atlas of grid showing the large-area uniformity of confined ice. **d**, Cryo-ET reconstruction reveals the thickness of confined ice (~30 nm) and the spatial distribution of embedded protein particles. Each blue circle represents an individual 20S proteasome particle. **e**, Representative cryo-ET slices extracted from **d** showing the top graphene–water interface, in-ice layer and bottom graphene–water interface, respectively. **f**, Comparative ice thickness statistics: confined ice in graphene reservoir sandwich displays a narrow peak at 30 nm, whereas conventional sandwiched ice exhibits a broader distribution

(52 ± 18 nm). **g**, Encapsulation efficiency comparison: confined ice in graphene reservoir sandwich achieves a high success rate of $91 \pm 6\%$ ($n = 2,219$ grid squares, eight repeated grids) versus conventional sandwiched ice at $67 \pm 4\%$ ($n = 1,973$ grid squares, three repeated grids). **h**, Particle motion versus accumulated electron dose in reservoir confined ice and conventional sandwiched ice. Each dotted line represents the motion average of thousands of individual particles. We performed three repeated measurements (dotted lines) for both reservoir confined ice and conventional sandwiched ice, using 12,035, 11,673 and 10,531 particles for reservoir confined ice, and 12,786, 13,073 and 11,656 particles for conventional sandwiched ice, respectively. Each solid line represents the mean of the corresponding dotted lines, and the error bars are s.d.

the particle movements in graphene sandwiches are lower than those on graphene (Fig. 4h and Supplementary Fig. 17) due to the enhanced mechanical properties of graphene sandwiches which can effectively delay the buckling of ice layer^{10,40}. Thus, our method improves both specimen preparation efficiency and ultimate image quality.

Uniform confined ice for high-resolution reconstruction

With precisely controlled thickness of confined ice, we successfully resolved structures of multiple macromolecules at near-atomic

resolution (Supplementary Table 1). We used the graphene reservoir to resolve structures of the LL-LtrB group II intron (RNA), demonstrating its capability for high-resolution single-particle cryo-EM analysis. The RNA particles exhibited high contrast and well-distributed orientations in the confined ice prepared by graphene reservoirs (Fig. 5a), enabling a high-resolution 3D reconstruction at 2.7 Å (Fig. 5b and Extended Data Figs. 6a and 7). The resulting map revealed distinct side chains with well-fitted atomic structures (Fig. 5c). In contrast, most RNA particles prepared on conventional grids either failed to enter the holes

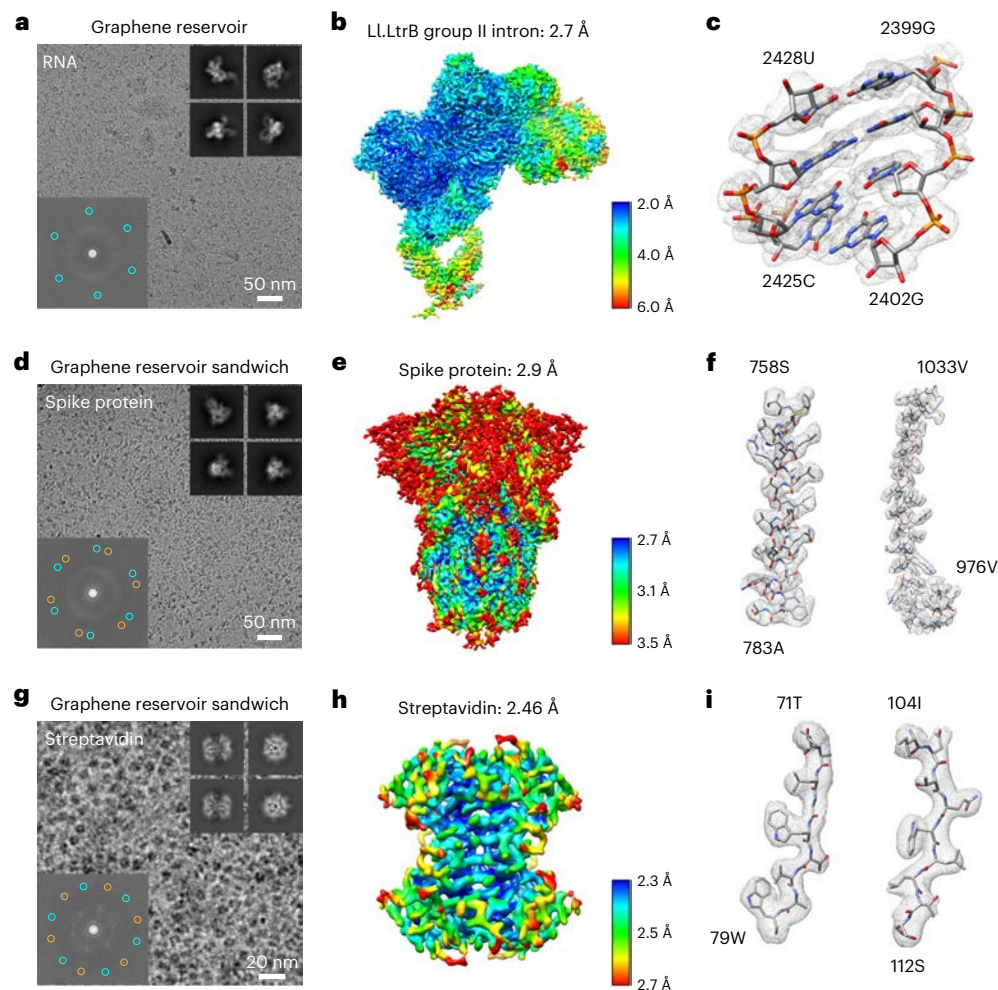


Fig. 5 | Uniform confined ice enabled high-resolution cryo-EM reconstructions. a,d,g. Representative cryo-EM images of RNA (LI.LtrB group II intron) in graphene reservoirs (a), spike protein in graphene reservoir sandwiches (d) and streptavidin in graphene reservoir sandwiches (g), respectively. Insets show typical 2D class averages (upper-right) and Fourier transform of the cryo-EM image (bottom left). The hexagonal diffraction spot patterns exhibit a twist angle in d and g, indicating the coexistence of top and

bottom graphene layers in the sandwich structure. **b,e,h.** The 3D density maps of LI.LtrB group II intron at 2.7 Å resolution (b), spike protein at 2.9 Å resolution (e) and streptavidin at 2.46 Å resolution (h), respectively. These maps are colored to visualize the local resolution distribution. **c,f,i.** Selected densities from the LI.LtrB group II intron map (c), spike protein map (f) and streptavidin map (i) with the corresponding atomic models docked.

or became aggregated (Extended Data Figs. 8a–c), both of which are common issues in RNA cryo-EM sample preparation⁴¹. These limitations lead to failed reconstruction of LI.LtrB group II intron. Beyond conventional cryo-EM, we further demonstrated the versatility of the graphene reservoir by extending it to graphene sandwich specimens. The SARS-CoV-2 spike protein particles encapsulated in graphene reservoir sandwiches also exhibited high signal-to-noise ratio (Fig. 5d). The particles were mono-dispersed and showed multiple orientations (Fig. 5d). The final 3D reconstruction was obtained at a near-atomic resolution of 2.9 Å (Fig. 5e,f and Extended Data Fig. 6b).

Many small proteins with molecular weight less than 100 kDa are often targets in drug discovery, but their cryo-EM reconstructions require a uniform thin ice to minimize the background noise given their low contrast in cryo-EM. In standard sample preparation, the heterogeneous ice thickness often limits the imaging of small proteins such as the 52-kDa streptavidin: excessively thin ice induces protein adsorption to the air–water interface and severe preferred orientation, whereas thicker ice not only fails to eliminate interfacial effects but also increases background noise (Extended Data Figs. 8d–f). In conventional sandwiched ice, the ice layer is typically too thick, rendering particle images nearly indistinguishable and preventing meaningful

3D reconstruction (Supplementary Fig. 18). In contrast to these limitations, graphene reservoir-based sandwiched ice enables clear visualization of individual streptavidin particles with a high signal-to-noise ratio under cryo-EM (Fig. 5g). Such a specimen easily yields a 3D reconstruction of streptavidin at 2.46 Å (Fig. 5h, Extended Data Fig. 6c and Supplementary Fig. 19), enabling the distinct identification of side chains of amino acid residues (Fig. 5i). The successful structural determination of streptavidin in this work reveals the great potential of precisely controlled confined ice in high-resolution cryo-EM by graphene reservoir sandwich.

Discussion

In conventional preparation of thin vitreous ice, the intrinsic instability of sub-100-nm liquid film significantly impedes the ice thickness control due to the energy-driven spontaneous thinning before vitrification^{16–18}. Notably, our graphene reservoir design inverts the paradigm. We demonstrated that careful design and fabrication of the perforated supporting film, with optimal hole dimensions and edge shapes, are crucial for creating graphene reservoirs with the appropriate sagging depth to retain the liquid film before vitrification. The spontaneous migration of liquid into graphene reservoirs ensures their exceptional liquid

retention capacity even under extreme blotting conditions, making them ideally suited for cryo-EM applications. Crucially, the preserved film thickness closely corresponds to the reservoir sagging depth. Benefiting from this, the long-pursued precise control of ice thickness in cryo-EM has been achieved by confining the liquid within graphene reservoirs with controllable depths. The obtained ice layers exhibit large-area uniformity and robust reproducibility.

Notably, our approach yields sandwiched ice with much greater thickness homogeneity, higher efficiency, and better reproducibility compared to conventional techniques, hereby addressing a key limitation of graphene sandwich techniques. Based on precisely controlled ice thickness, we have successfully determined the 3D atomic structures of multiple macromolecules, including sub-100-kDa small biomolecules encapsulated in graphene sandwiches. Moreover, we anticipate that the graphene reservoir sandwich will also benefit many emerging imaging techniques, from laser flash melting of cryo-EM samples for overcoming preferred orientation⁴² and investigating biomolecular dynamics⁴³ to in situ liquid electron microscopy, thereby paving the way for studying biological or chemical structures in their natively hydrated state and resolving their dynamics in liquid.

Our approach provides a robust strategy for generating uniform thin liquid films and can also be applicable to spray-based methods. In conventional methods, sprayed droplets are typically thick (Extended Data Figs. 9a–d and Supplementary Videos 3 and 4), restricting imaging only to peripheral regions where the vitrified ice is sufficiently thin. The greatest hurdle in spray methods is the spreading of the liquid on the grid¹⁵, and previous attempts show that self-wicking nanowire grids promote liquid spreading through capillary suction^{44,45}, enabling the formation of thin ice for data acquisition. Consistent with this principle, we show that liquid can spontaneously spread into the graphene reservoirs after droplet deposition on the grid surface, contributing to uniform ice layers within graphene reservoirs (Extended Data Figs. 9e,f and Supplementary Video 5) and potentially providing a promising strategy for improved ice thickness control in time-resolved cryo-EM^{8,9}. Beyond electron microscopy, the ability to precisely manipulate confined ice or liquid provides a powerful tool for investigating the emergent properties of confined systems related to various physical, chemical and biological processes^{22,46}.

Online content

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Methods

Preparing uniform holes with industry-compatible lithography

Substrate. Before performing lithography, a 100-nm-thick Cu film was deposited by magnetron sputtering onto a 6-inch silicon wafer to act as the substrate. Deposition occurred at a basal pressure of ~0.08 Pa and a radio frequency power of ~100 W.

Lithography. Lithographic techniques are widely employed to fabricate templates for grids featuring periodic hole arrays, such as the commonly used Quantifoil and C-flat grids^{35,47}. To ensure exposure precision, a 90-nm-thick BARC (bottom anti-reflective coating, AR3-600) layer was first spun onto the Cu film/silicon wafer and baked at 180 °C for 1 min. Next, a positive photoresist (DK3030B or DK1150A, Red Avenue New Materials Group Co. or UV135G, Dow) was spin-coated and soft-baked at 120 °C for 1 min. The wafer was then exposed to 248 nm deep ultraviolet light in a lithography tool (Canon, 3030ex6) to transfer the predesigned hole patterns from the mask to the photoresist. A post-exposure bake at 130 °C for 1 min followed, inducing a solubility change in the exposed regions. Finally, the hole patterns were obtained by selectively dissolving the exposed photoresist using a development process. Notably, this lithography process can be readily implemented in both standard university microfabrication facilities and industrial fabrication environments (for example Atomface New Materials (Lianyungang) Co.; <https://www.atomface-tech.com>).

Fabricating holey carbon grids

Etching. To release the holey photoresist film from the silicon wafer substrate, both the BARC layer and Cu film were selectively etched. The BARC layer was removed via O₂ reactive ion etching at 200 W for 50 s. The wafer was then immersed in a 1 M (NH₄)₂S₂O₈ aqueous solution to dissolve the Cu film. After etching, the photoresist film remained attached to the underlying silicon wafer substrate. The photoresist film/silicon wafer was lifted from the etchant at an angle of ~45° and subsequently transferred to the water surface. The photoresist film was then slowly released onto the water surface at the same ~45° angle, ultimately resulting in the film floating on the water surface. Details are shown in Supplementary Video 1.

Transfer. The transfer process involved (1) immersing a filter paper in deionized water; (2) positioning metal grids (for example, Au 300 mesh) on the submerged paper; then (3) depositing the holey photoresist film onto the metal grids by gradually siphoning off the water. Finally, these grids were air-dried to ensure reliable adhesion between holey photoresist film and metal grids.

Carbon deposition. Using the holey photoresist film as a template, we successfully replicated its morphology by depositing a 30-nm-thick carbon film onto the photoresist-coated grids via pulsed-current sputtering (40–70 A, Quorum Q160T ES), thus fabricating a holey carbon film with identical pore architecture.

Rinsing and drying. The grids were submerged in acetone to remove the photoresist template and then transferred into isopropanol (IPA) for further rinsing. After drying, the homemade holey carbon grids were obtained.

Graphene transfer for preparing graphene reservoirs

Graphene raw materials. The monolayer and trilayer graphene films grown on the copper foils were purchased from Beijing Graphene Institute. The trilayer graphene films, synthesized via the synchronous growth method⁴⁸, exhibited exceptional layer homogeneity, nearly 100% Bernal stacking and superior mechanical properties.

Graphene transfer. The graphene films were transferred onto the homemade holey carbon grids using a polymer-free transfer method³⁷.

First, the stearic acid solution (0.005 wt%, dissolved in IPA) was dropped onto the graphene surface. After the evaporation of IPA, self-assembled monolayer of stearic acid spontaneously formed on the graphene. The self-assembled monolayers could maintain the intactness of freestanding graphene on the etchant solution after the copper foil was etched off, thus circumventing the use of polymer in conventional transfer methods. Subsequently, homemade holey carbon grids were carefully placed onto the surface of freestanding graphene. The vdW interactions between graphene membrane and holey carbon film ensured the reliable adhesion, contributing to the high-coverage graphene reservoirs.

Rinsing and drying. The graphene grids were transferred onto the deionized water surface to remove away the residual etchant, followed by rinsing in IPA for 2–5 min to eliminate the stearic acid molecules. After being taken out from the IPA, the graphene grids were air-dried.

We note that the fabrication procedures for conventional graphene grids are nearly identical to those for graphene reservoir grids, differing only in the type of grid used. Conventional graphene grids were prepared using the commonly used commercially available grids (Quantifoil-R1.2/1.3).

AFM, SEM and scanning transmission electron microscopy characterizations

The morphologies of graphene reservoirs were characterized by AFM (Bruker Dimension Icon equipped with Nanoscope V controller) in tapping mode using a commercially available AFM tip (OTESPA-R3, Bruker). The coverage of the graphene reservoir grid was collected using SEM (2 kV, Hitachi S-4800), and the aberration-corrected scanning transmission electron microscopy images of graphene were carried out using a Nion U-HERMS200 microscope at 60 kV.

Preparation of biological samples

The 20S proteasome was expressed and purified from *Escherichia coli* cells⁴⁹. In brief, a plasmid containing the α and β subunit of the 20S proteasome was constructed, and the amino terminus of the β subunit was His-tagged. Then the plasmid was transformed into *E. coli* cells and left overnight for expression. Cells lysates were obtained by centrifuging and sonicating the cells, which were finally loaded onto a nickel column (GE Health-care) for affinity purification, the final elution was diluted to 0.3 mg ml⁻¹ for use. For LI.LtrB group II intron⁵⁰, the DNA template for in vitro transcription was generated by PCR, the sense strand of the amplicon contained an upstream sequence, a T7 promoter and the gene encoding the LI.LtrB group II intron from 5' to 3'. Then the RNA was synthesized by in vitro transcription using T7 RNA polymerase and DNA template, which was incubated at 37 °C overnight. To remove the template DNA, 1 μ l of DNase I (2 U μ l⁻¹) was added and incubated at 37 °C for 1 h. The transcription mixture was purified by size-exclusion chromatography under native conditions. A Superose 6 Increase 5/150 GL column (column volume ~3 ml) was selected, which was equilibrated with non-denaturing buffer (20 mM HEPES-KOH, 150 mM KCl and 10 mM MgCl₂). Fractions containing the full length of LI.LtrB group II intron were collected and diluted to 0.4 μ M for use. Apoferritin was purchased from Sigma-Aldrich (cat. no. A3660) and diluted to 1 mg ml⁻¹ for use. Streptavidin was purchased from NEB (cat. no. N7021S) and diluted to 0.1 mg ml⁻¹ for use. The SARS-CoV-2 spike protein was kindly provided by Q. Zhou at Westlake University and diluted to 0.4 mg ml⁻¹ for use.

Cryo-EM specimen preparation using graphene reservoirs

For cryo-EM specimens used to demonstrate ice thickness control, a 3 μ l sample solution was pipetted onto the monolayer graphene reservoir grid (Au 300 mesh, 1,200-nm hole diameter), which had been glow-discharged in a plasma cleaner machine (Harrick Plasma) for 15–30 s at a low power setting. The grid was transferred into a Vitrobot

Mark IV (Thermo Fisher Scientific) with the chamber temperature set to 8 °C and humidity maintained at 100%. The blotting force was set to −2, the waiting time to 10 s and different blot times (2 s, 6 s, 8 s, 10 s, 12 s, 15 s and 20 s) were tested. The grids were then plunge-frozen in liquid ethane and stored in liquid nitrogen for subsequent cryo-EM imaging.

For monolayer graphene reservoir grids with varying hole diameters, the Vitrobot blot times were 5–6 s for 600-nm, 7–8 s for 800-nm, 8–10 s for 1,200-nm and 12–13 s for 2,000-nm diameters. All other parameters remained as described above. Different blot time for grids with varying hole diameters were also tested as follows: 2 s, 4 s, 6 s and 8 s for 600 nm; and 2 s, 6 s, 12 s and 16 s for 2,000 nm. For structural determination of the LL.LtrB group II intron, we used a monolayer graphene reservoir grid (Au 300 mesh, 1,200-nm hole diameter) with a blot time of 10 s, maintaining all the other parameters as described above.

For cryo-EM specimens prepared by ESI-cryoPrep, Quantifoil grids, Quantifoil grids with graphene and graphene reservoir grids were used, with the same glow-discharging parameters as mentioned above. The other parameters were kept the same as previous methods⁵¹: the ESI needle was connected to the positive electrode of a DC power supply (+3 kV), 10 µl of a sample solution (apoferritin diluted with 1× PBS to a concentration of 1 mg ml^{−1}) was injected using a precise and stable automatic injection device, and the negative electrode was connected to the tweezer holding the grid. The liquid droplets reached the grid at a moderate injection flow rate (100–300 nl min^{−1}) within tens of seconds. Then the grid was manually frozen in nitrogen-cooled liquid ethane for cryo-EM analysis.

Fabrication of cryo-EM samples embedded in graphene reservoir sandwiches

Preparation of top graphene for sandwich structure. (1) We used the trilayer graphene as the top layer of sandwich structure. The commercially available trilayer graphene (Beijing Graphene Institute) was grown on copper foil. To remove the graphene on the backside of the copper foil, the foil was put into a reactive ion etcher (Pico SLS, Diener) with an air flow of 5 sccm. The air plasma was then generated at a controlled power of 150 W. After a 3-min treatment, the graphene on the backside was completely removed, while the trilayer graphene on the opposite side of the copper foil remained intact.

(2) The trilayer graphene/copper foil was placed on the surface of 1 M (NH₄)₂S₂O₈ aqueous solution to etch the Cu foil. After etching, the freestanding trilayer graphene remained floating on the etchant solution surface. Notably, a self-assembled monolayer of stearic acid could be used to increase the robustness of freestanding trilayer graphene if needed. The fabrication procedures for the self-assembled monolayers are detailed in the ‘Graphene transfer for preparing graphene reservoirs’ section.

(3) The freestanding trilayer graphene was carefully transferred from the etchant solution onto the surface of deionized water using a sieve mesh. This transfer process onto fresh deionized water was repeated twice to ensure the removal of residual etchant. Subsequently, the freestanding trilayer graphene was transferred onto the surface of the buffer solution.

Preparation of bottom graphene for sandwich structure. (4) A 3-µl sample solution was pipetted onto the monolayer graphene reservoir grid (Au 300 mesh, 1,200-nm hole diameter), which had been glow-discharged in a plasma cleaner machine (Harrick Plasma) for 30 s at a low power setting.

(5) Then the sample-loaded grid was immediately immersed beneath the freestanding trilayer graphene film and then the film was scooped up rapidly.

Blotting and plunge-freezing. (6) The graphene reservoir sandwich grid was transferred into a Vitrobot Mark IV (Thermo Fisher Scientific), and the temperature of the Vitrobot chamber was set as 8 °C with a

humidity of 100%. The blotting force was set to −1, the waiting time was set to 10 s and the blot time was set to 20–22 s.

(7) The grid was subsequently plunged into liquid ethane and stored in liquid nitrogen for further cryo-EM imaging.

For the trilayer graphene reservoir grids, all the parameters were kept the same.

Conventional cryo-EM specimen preparation

Conventional holey carbon grids (Quantifoil, Au 300 mesh, R1.2/1.3) were used for comparing particle behavior between conventional grids and graphene reservoir or graphene reservoir sandwich grids. The sample solution (4 µM for LL.LtrB group II intron or 1 mg ml^{−1} for Streptavidin) was applied to grid using a 3-s blot time, with all other parameters consistent with graphene reservoir protocols.

Conventional graphene grids (Quantifoil, Au 300 mesh, R1.2/1.3) featuring protruding holes served as controls. For cryo-EM specimen preparation, apoferritin samples (3 µl) were applied to these grids using a 10-s blot time, with all other parameters consistent with graphene reservoir protocols.

For conventional graphene sandwiched sample, a 3-µl sample solution was pipetted onto the conventional graphene grid, which had been glow-discharged in a plasma cleaner machine (Harrick Plasma) for 30 s at a low power setting. Then the sample-loaded grid was also immediately immersed beneath the freestanding trilayer graphene film and then scooped the film up rapidly. Next, the conventional graphene sandwich grid was positioned on the filter paper (Ted Pella, 47000-100), which was used to remove excess solution from the sandwiched area. Finally, the manually blotted grid was plunged into liquid ethane and stored in liquid nitrogen for further cryo-EM imaging.

Cryo-EM data collection and structural analysis

The single-particle Cryo-EM dataset of graphene reservoir for LL.LtrB group II intron was automatically collected using EPU software on a Thermo Fisher Scientific Titan Krios G3i TEM (300 kV), equipped with a GIF-Quantum energy filter (20 eV) and a Gatan K3 direct electron detector. A Total of 6,528 micrographs were collected with a pixel size of 0.83 Å. For the dataset of conventional graphene grid, graphene reservoir grid (600 nm, 1,200 nm and 2,000 nm), conventional graphene sandwich grid and graphene reservoir sandwich grid for motion analysis, 220, 253, 180, 191, 180 and 103 micrographs were automatically collected, respectively, with a pixel size of 0.97 Å, using SerialEM software⁵² on an FEI Titan Krios TEM (300 kV) equipped with a Gatan K3 direct electron detector. For the dataset of graphene reservoir sandwich grid for SARS-CoV-2 spike protein and streptavidin, 4,377 micrographs with a pixel size of 0.8374 Å, and 3,646 micrographs with a pixel size of 0.5141 Å were automatically collected, respectively, using AutoEMation software⁵³ on a Thermo Fisher Scientific Titan Krios TEM (300 kV) equipped with a Gatan K3 direct electron detector and a GIF-Quantum energy filter (20 eV). For the dataset of graphene reservoir grid and conventional graphene grid for apoferritin, 178 and 304 micrographs with a pixel size of 0.836 Å were automatically collected using SerialEM software on an FEI Tecnai Arctica TEM (200 kV) equipped with a Gatan K2 camera.

For the dataset of conventional grid without graphene for streptavidin, 1,153 micrographs with a pixel size of 0.5141 Å were automatically collected, respectively, using AutoEMation software⁵³ on a Thermo Fisher Scientific Titan Krios TEM (300 kV) equipped with a Gatan K3 direct electron detector and a GIF-Quantum energy filter (20 eV).

All the micrographs aforementioned were fractioned to 32 frames with a summed electron dose of 50 e[−]/Å² and motion-corrected by MotionCor2 (ref. 54). Then the motion-corrected micrographs were imported into CryoSPARC⁵⁵ for patch contrast transfer function (CTF) estimation, micrograph selection, particle picking, 2D classification, hetero refinement and the final 3D reconstruction. The number of particles used for the final 3D reconstruction of graphene reservoir grid for LL.LtrB group II intron, graphene reservoir sandwich grid for

SARS-CoV-2 spike protein and streptavidin were 179,409, 150,268 and 106,422, respectively. The reported resolutions were 2.70 Å, 2.90 Å and 2.46 Å, respectively, determined using Fourier shell correlation = 0.143 as the cutoff criterion. Protein Data Bank 7D0G, 6X29 and 8HRM were used to dock into the density maps of LI.LtrB group II intron (EMD-66426), SARS-CoV-2 spike protein (EMD-66423), and streptavidin (EMD-66424), and then real-space refined using CryoNet.Refine⁵⁶. All the structure details were visualized using Chimera⁵⁷. All these cryo-EM data collection, refinement and validation statistics are summarized in Supplementary Table 1. The Euler angles for each particle in the final particle dataset of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin were plotted for the orientational distribution analysis.

To analyze the spatial distribution of protein particles in within graphene reservoirs, local CTF refinement in CryoSPARC was performed to estimate the local defocus values of individual apoferritin particles from the graphene reservoir grid, and subsequently to plot the 3D graphics according to their *x*, *y* and *z* coordinates, showing the spatial distribution of inclined planes relative to the *x*-*y* plane⁵¹.

In our particle utilization analysis comparing conventional graphene grids with graphene reservoir grids (10-s blot time), we initially selected 89,151 and 34,407 raw particles, respectively. Following refinement, 33,900 (38.0% utilization) and 24,028 (69.8% utilization) particles remained for final 3D reconstruction.

To measure particle motion as a function of accumulated electron dose in conventional graphene grids, graphene reservoir grids with different hole diameters, conventional graphene sandwich grids and graphene reservoir sandwich grids, Bayesian polishing in Relion 3.0.8 was used to determine the particle coordinates in each frame, which were then used to calculate the root-mean-squared displacements of per particle and plotted versus electron dose. For each sample, three particle-motion measurements were performed, and each of them was based on thousands of particles.

Cryo-ET data collection and analysis

The cryo-ET data were collected with tilt angles ranging from +60° to -60° at 3° intervals, using SerialEM software equipped with PACETom³⁸ on a Thermo Fisher Scientific Titan Krios TEM (300 kV) equipped with a Gatan K3 direct electron detector and a GIF-Quantum energy filter (20 eV), or using SerialEM software on an FEI Titan Krios TEM (300 kV) equipped with a Gatan K3 direct electron detector. The pixel sizes were 0.8697 Å (graphene reservoir grid with 600-nm hole), 1.642 Å (graphene reservoir grids with 800-nm, 1,200-nm and 2,000-nm holes and graphene sandwich grid), and 1.25 Å (graphene reservoir grids with 600-nm, 1,200-nm, 2,000-nm holes and graphene sandwich grid). Each tilt was fractionated into eight frames with a cumulative dose of 3 e⁻/Å² per tilt, resulting in a total dose of 120 e⁻/Å² for the complete tilt series. The tilt series were firstly motion-corrected using MotionCor2 and then imported into IMOD⁵⁹ for further alignment and tomogram reconstruction. The final tomograms were generated with a binning factor of 4. The position of 20S proteasome particles in the graphene reservoir sandwiched sample was manually identified and plotted⁶⁰.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The cryo-EM density maps and coordinates of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin have been deposited into EMDB under accession numbers EMD-66426, EMD-66423 and EMD-66424, and into the Protein Data Bank under accession numbers 26HH, 26DA and 26CW, respectively. The raw data of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin have been deposited into EMPIAR, with the corresponding accession codes EMPIAR-13520, EMPIAR-13521 and EMPIAR-13519, respectively. Any additional data supporting the findings

in this manuscript are available from the corresponding authors upon reasonable request. Source data are provided with this paper.

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Author contributions

L.Z., H.-W.W. and H.P. conceived the idea. L.Z. and X.Z. prepared the graphene reservoir grids. J.S., N.L., J.X., Z.W., C.Z., J.Y. and Y.X. carried out the cryo-EM characterizations. N.L., J.S., L.Z. and H.-W.W. analyzed the cryo-EM data. J.C., L.Z., Z.D. and X.W. performed theoretical simulations and mechanical analyses. X.Z., L.Z., X.G. and W.S. carried out the characterizations of graphene reservoirs. L.Z. and H.L. contributed to the lithography. B.C. and L.S. synthesized graphene film. L.Z., N.L., H.P. and H.-W.W. wrote the paper with contributions

from J.S., X.Z., J.C., Z.D. and X.W. All authors discussed the results and gave approval to the final version of the paper.

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Competing interests

The authors H.P., H.-W.W., L.Z., J.S. and X.Z. have filed Chinese patent applications (ZL202410883073.3, PCT/CN2026/093217 and 202610189344.4), covering the procedures of preparing TEM grids

for fabricating graphene reservoirs. All other authors declare no competing interests.

Additional information

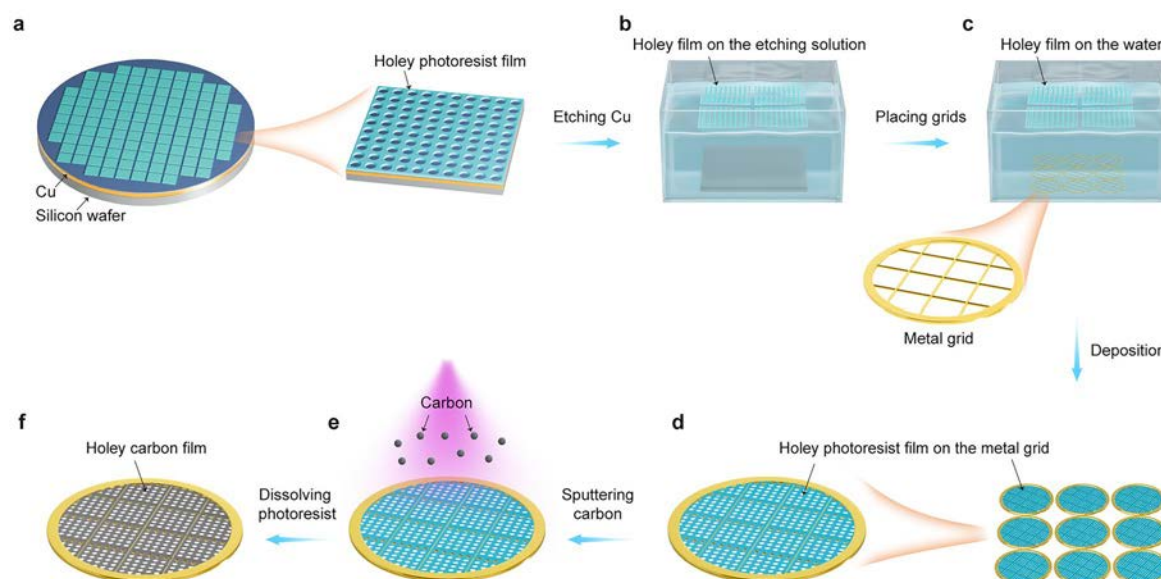
Extended data is available for this paper at <https://doi.org/10.1038/s41592-026-03244-1>.

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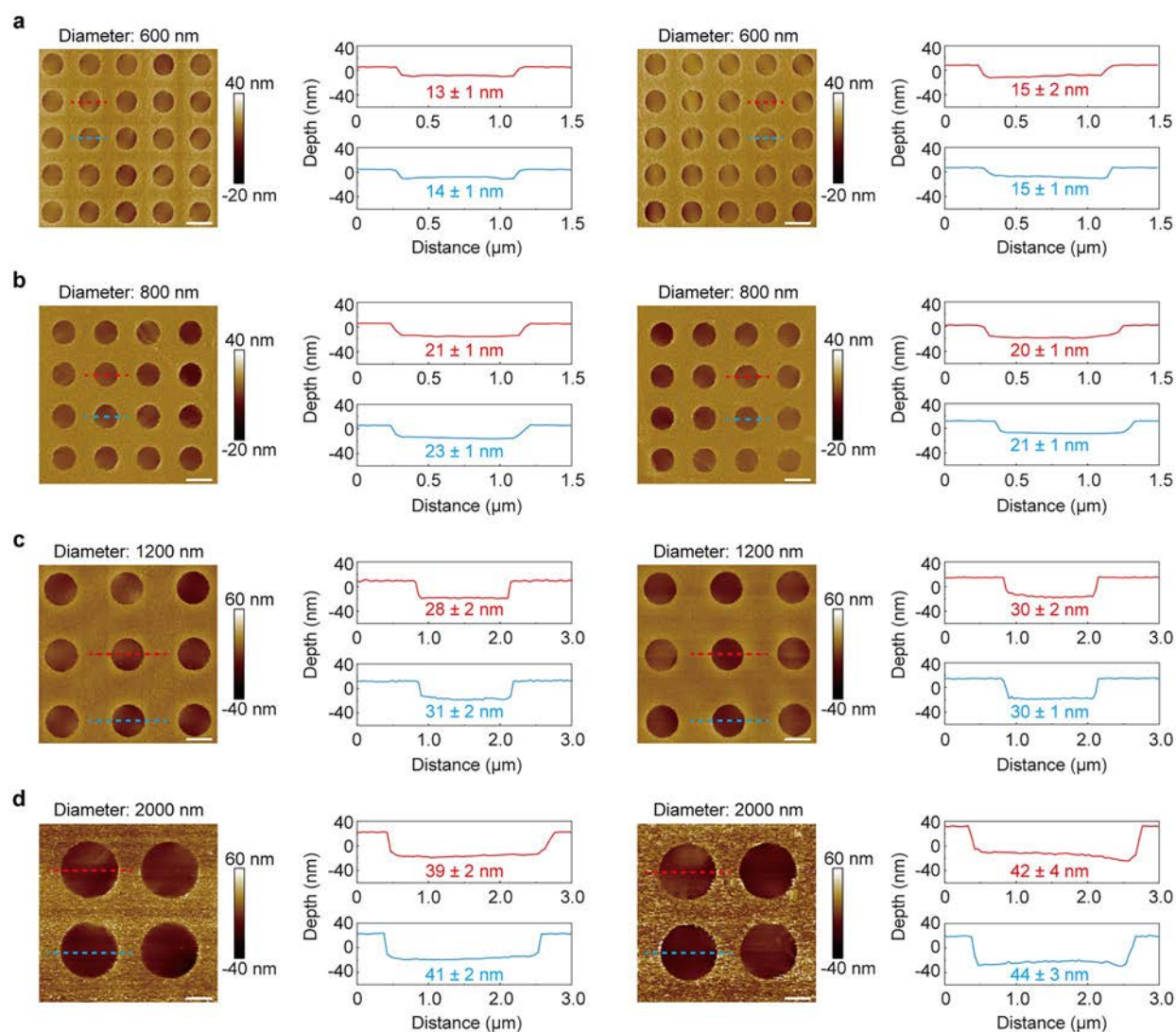
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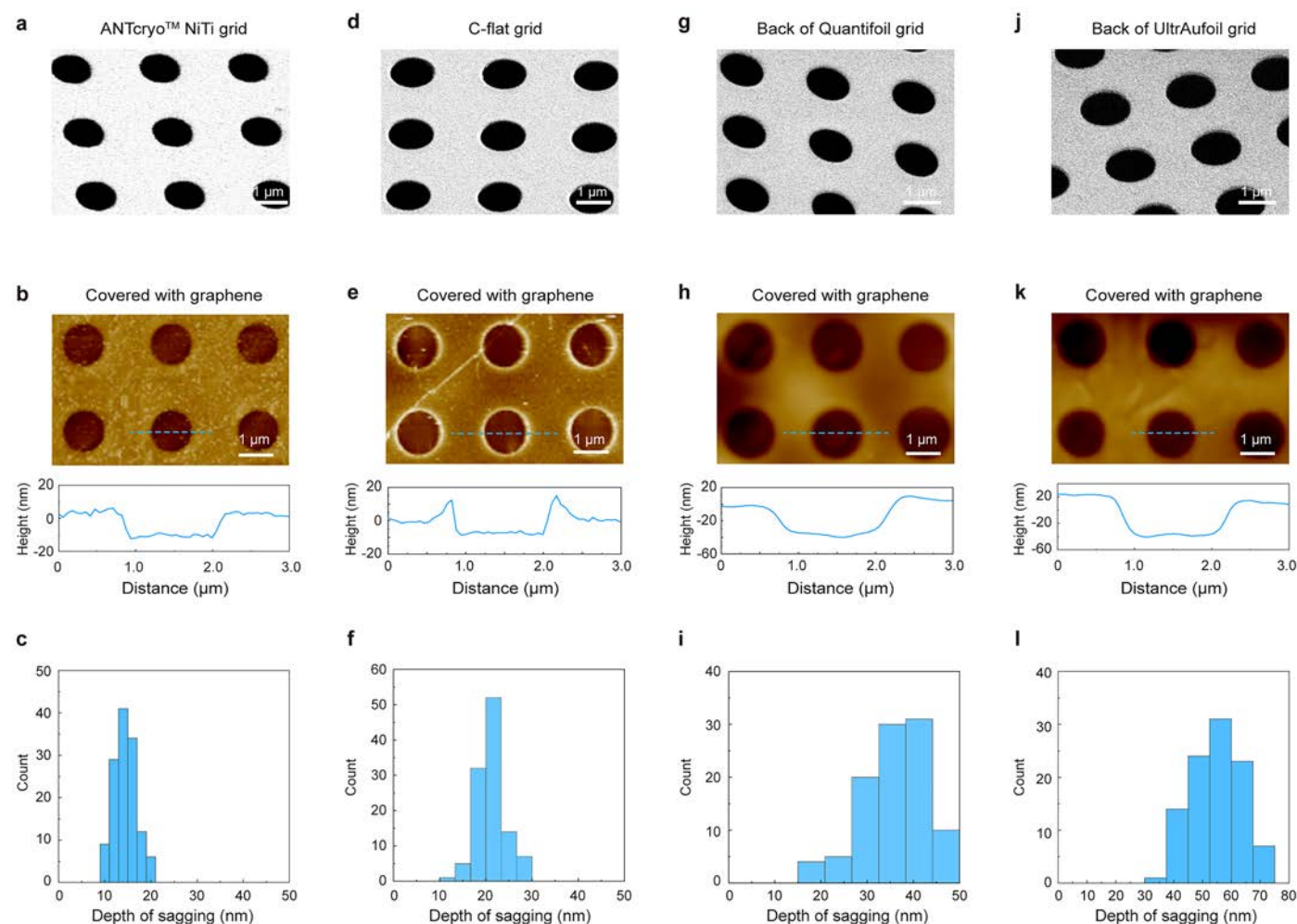


Extended Data Fig. 1 | Schematic illustration of scalable fabrication of homemade EM grids. **a**, Lithography enables fabrication of holey photoresist film on a Cu/silicon wafer. **b**, The Cu film is etched to detach the holey photoresist film from the silicon substrate. After etching, the holey photoresist film is floated off on the etching solution. **c**, The holey photoresist film is transferred onto the water surface, followed by placing metal grids beneath the floating film. **d**, The

holey photoresist film is deposited onto the metal grids. After drying, the holey photoresist film is tightly attached on the grids. **e**, Carbon is sputtered onto the holey photoresist template, which replicates the smooth, rounded morphology of photoresist hole edges. **f**, The holey photoresist template is dissolved in acetone solution, leaving the holey carbon film adhered to the metal grid, thus forming the homemade holey carbon grids.

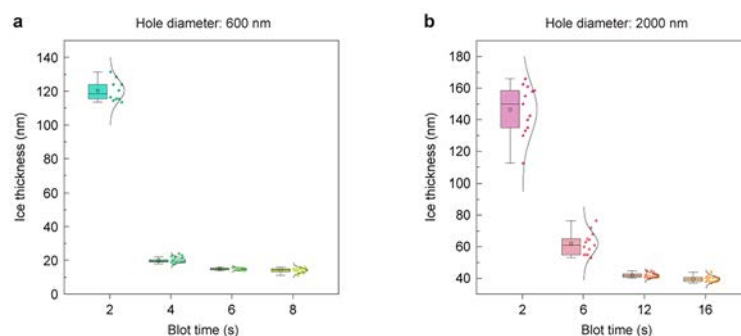


Extended Data Fig. 2 | Depth modulation of graphene reservoir by controlling the diameter of hole. a-d, Typical AFM images and depth profiles of graphene reservoirs with hole diameters of 600 nm (a), 800 nm (b), 1,200 nm (c), and 2,000 nm (d). More than 100 holes were measured independently with similar results.



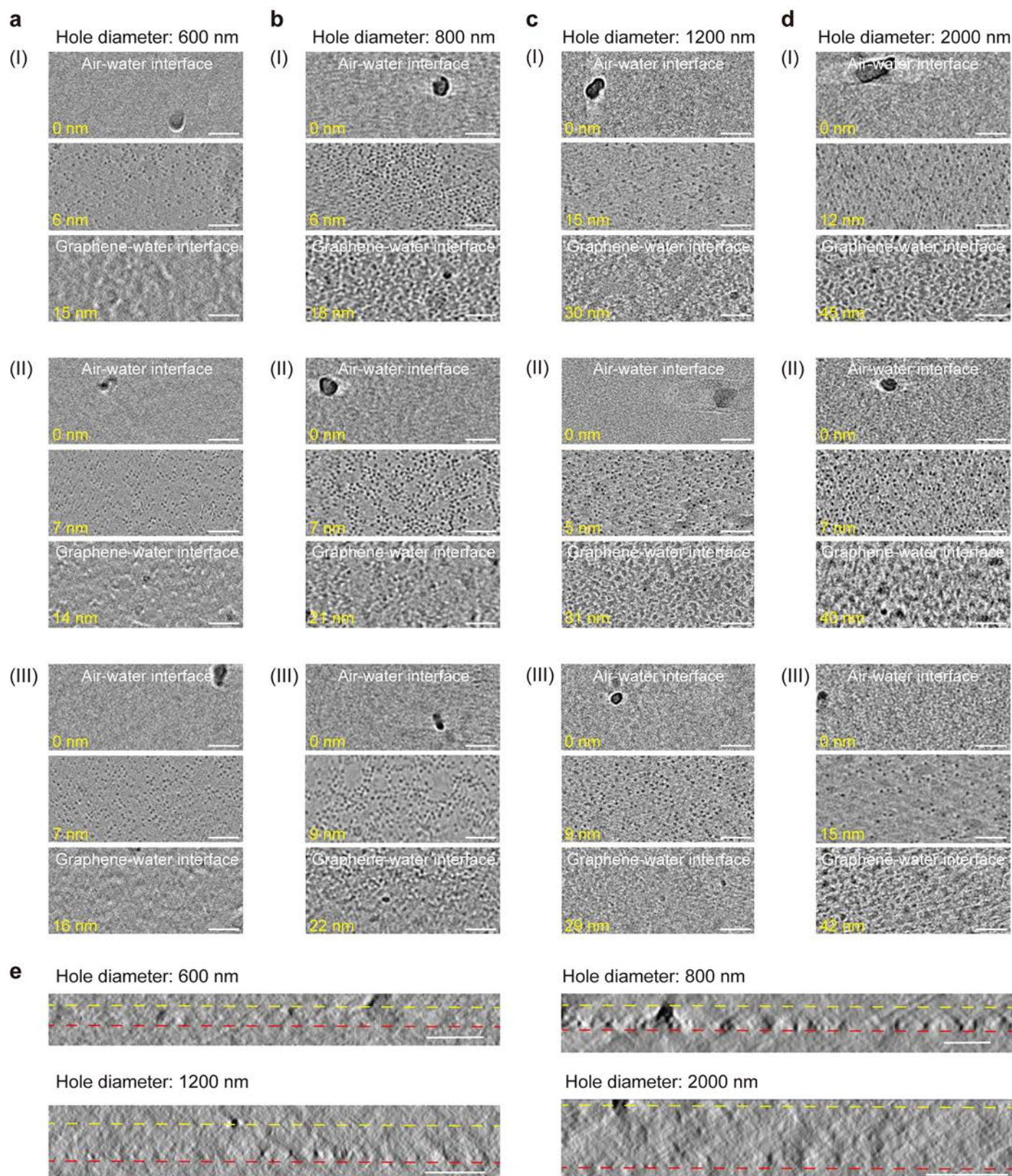
Extended Data Fig. 3 | Robust compatibility of the confinement strategy with existing grid technologies. a, d, g, j. Tilted SEM images showing the smooth hole edges of commercially available ANTcryo™ grids (a), C-flat grid (d), backside of Quantifoil grid (g) and backside of UltrAufoil grid (j), respectively. **b, e, h, k.** AFM images revealing the sagging morphology of graphene on the ANTcryo™ grids (b), C-flat grid (e), backside of Quantifoil grid (h) and backside of UltrAufoil grid

(k), respectively. The lower panels present height profiles measured along the dotted blue lines indicated in the upper AFM images. More than 100 holes were measured independently with similar results. **c, f, i, l.** Sagging depth distribution of graphene on the ANTcryo™ grids (c), C-flat grid (f), backside of Quantifoil grid (i) and backside of UltrAufoil grid (l), respectively.



Extended Data Fig. 4 | Dependence of ice layer thickness on blot time in graphene reservoirs with different hole diameters. a, For graphene reservoirs with a 600 nm hole diameter, the average ice thicknesses of different blot time are shown as follows: 2 s, 120 ± 6 nm ($n = 10$); 4 s, 20 ± 1 nm ($n = 28$); 6 s, 15 ± 1 nm ($n = 16$); and 8 s, 14 ± 1 nm ($n = 16$). The ice thickness stabilizes at approximately 15 nm after 6 s of blotting. **b,** For graphene reservoirs with a 2,000 nm hole diameter, the average ice thicknesses of different blot time are shown as

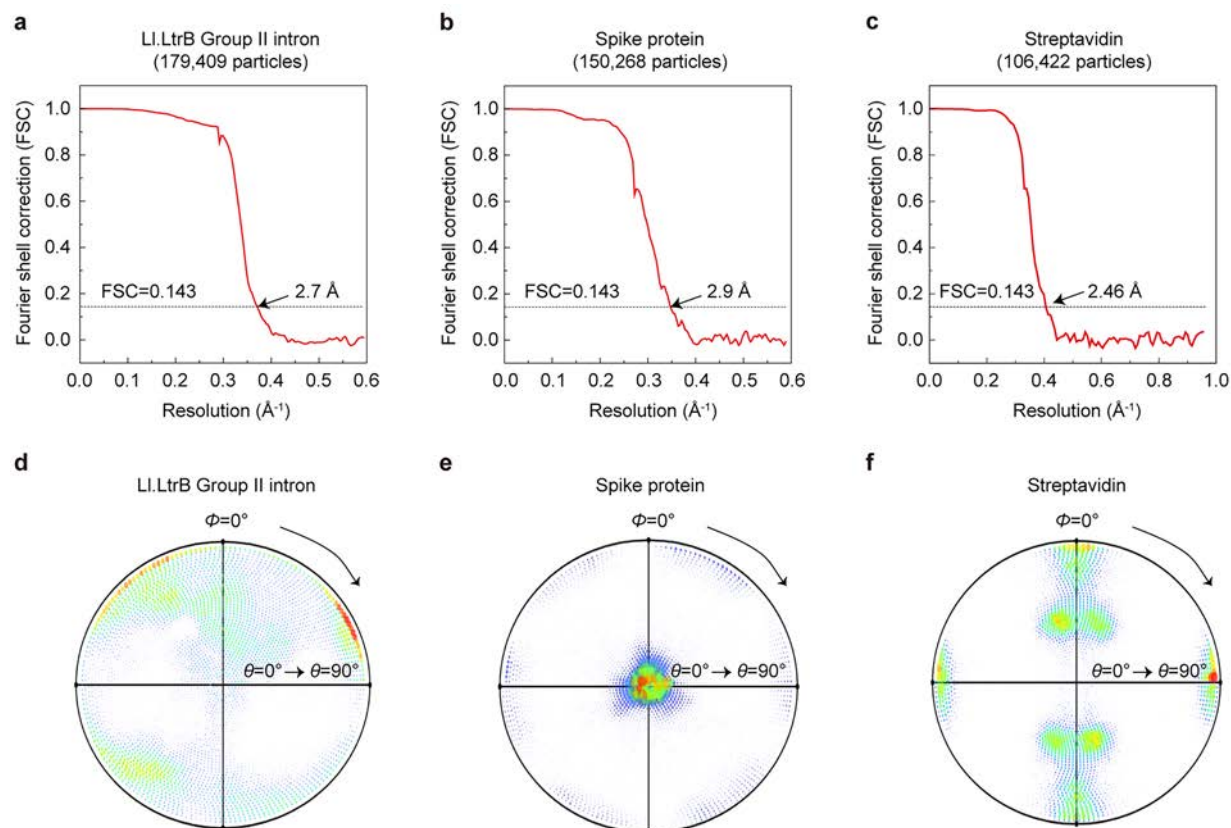
follows: 2 s, 146 ± 15 nm ($n = 13$); 6 s, 62 ± 7 nm ($n = 13$); 12 s, 42 ± 1 nm ($n = 23$); and 16 s, 40 ± 2 nm ($n = 16$). The ice thickness reaches ~40 nm after 12 s. For the box-and-raincloud plots in (a) and (b), the box limits indicate the first and third quartiles (25th and 75th percentiles), the center line indicates the median, the black square indicates the mean, the whiskers indicate the minimum and maximum values, individual data points are shown as jittered points, and the half-violin represents the kernel density estimate.



Extended Data Fig. 5 | See next page for caption.

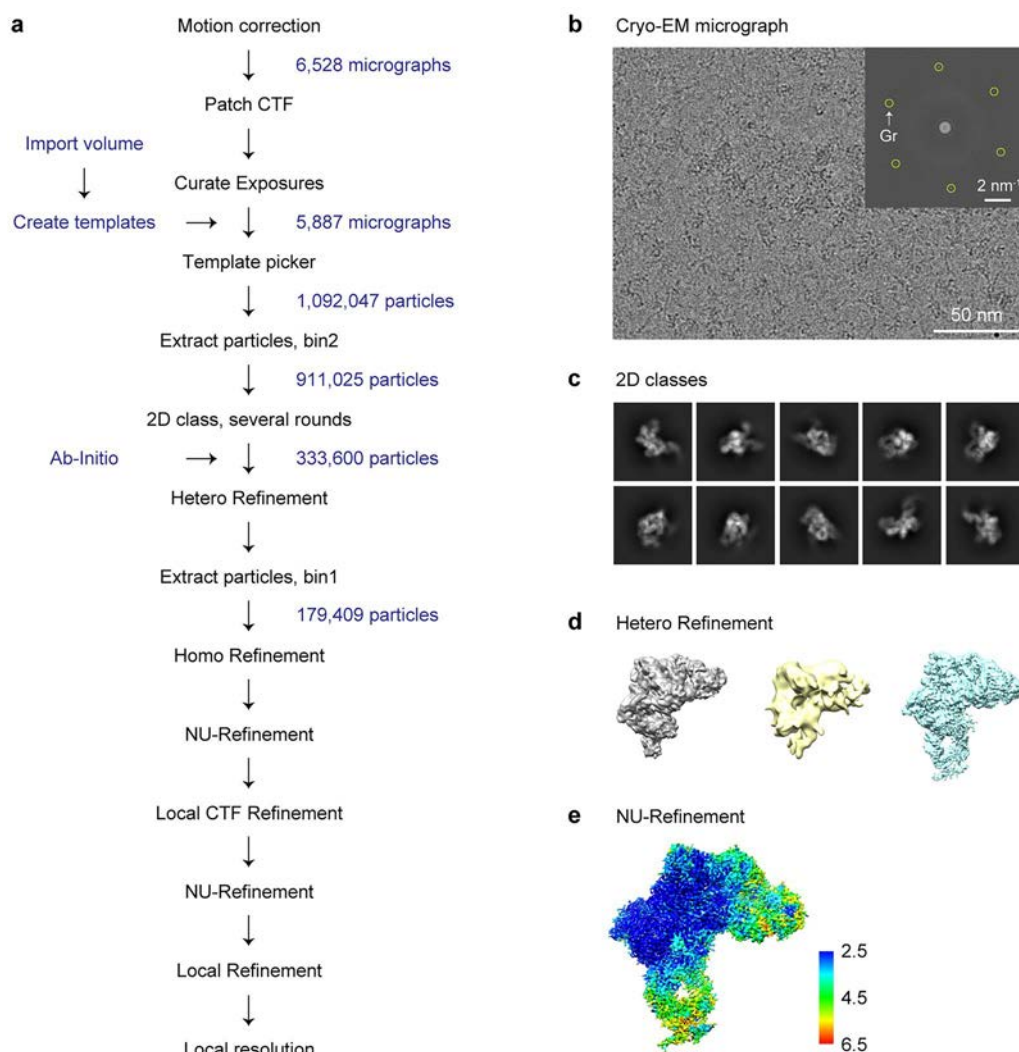
Extended Data Fig. 5 | Controllable ice thicknesses by varying the hole diameter of graphene reservoirs. **a–d**, Representative cryo-ET slices demonstrate controllable ice thicknesses of approximately 15 nm (**a**), 20 nm (**b**), 30 nm (**c**), and 40 nm (**d**), achieved using graphene reservoirs with hole diameters of 600 nm, 800 nm, 1,200 nm, and 2,000 nm, respectively. For each hole diameter, the graphene reservoirs produce uniform ice thickness across different specimen regions. Streptavidin

particles were clearly observed in the middle layers of (a), (b), (c) and (d). More than 10 tilt-series datasets were collected for each group independently with similar results. **e**, The X–Z slices showing the protein distribution in the graphene reservoir with a hole diameter of 600 nm, 800 nm, 1,200 nm, and 2,000 nm, respectively. The yellow and red dash lines indicate the air–water interface and graphene–water interface, respectively. Scale bars, 50 nm.



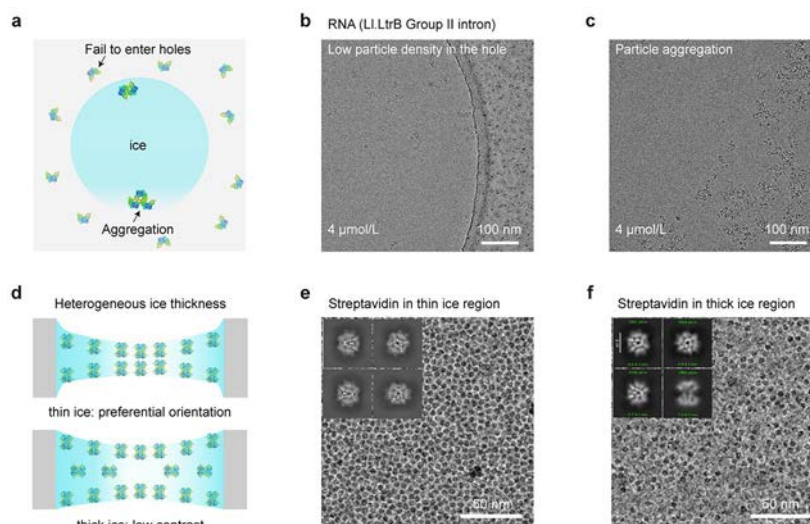
Extended Data Fig. 6 | Resolution and particle orientational distribution of the testing samples. a-c, The Gold-Standard Fourier Shell Correlation (FSC) curves of LI.LtrB Group II intron (a), spike protein (b), and streptavidin (c) reconstructions, respectively. FSC = 0.143 was used for calculating the

resolution. The number of particles for final 3D reconstruction was also listed. **d-f,** Particle orientational distributions showing the orientations of Group II intron LtrB (d), spike protein (e), and streptavidin (f), respectively.



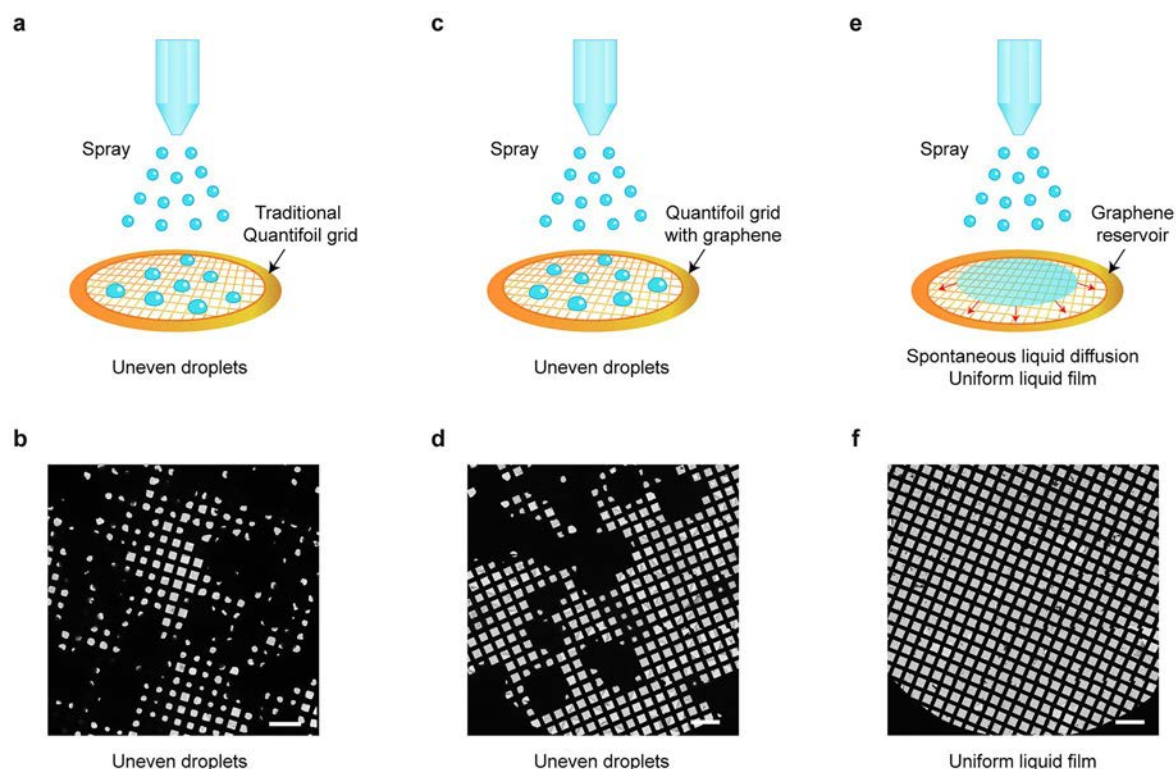
Extended Data Fig. 7 | Cryo-EM data-processing workflow of RNA (L1.LtrB Group II intron) within confined ice in graphene reservoirs. a, The 3D reconstruction workflow of L1.LtrB Group II intron. **b,** A representative cryo-EM image of L1.LtrB Group II intron on the graphene reservoirs. Inset: the Fast Fourier Transform (FFT) of the cryo-EM image, the hexagonal diffraction spot patterns indicate the existence

of a graphene layer. **c,** Representative 2D classification results of L1.LtrB Group II intron on the graphene reservoirs. **d-e,** The 3D classification results (**d**) and the final reconstruction map at a 2.7-Å resolution (**e**), respectively. The color bar in (**e**) indicates the local resolution distribution (Å).



Extended Data Fig. 8 | Cryo-EM imaging using the conventional sample preparation method without graphene. **a**, Schematic shows that most RNA (L1.LtrB group II intron) particles fail to enter the holes and tend to aggregate in the ice layer. **b**, A typical cryo-EM image shows low RNA particle density in the hole despite a high concentration of 4 μM , with most particles adsorbed on the holey carbon film. **c**, A typical cryo-EM image reveals the severe aggregation of RNA particles. **d**, Schematics illustrate that heterogeneous ice thickness and preferential orientation lead to the failed reconstruction of

streptavidin, when prepared on conventional grids without graphene. **e**, A representative cryo-EM image reveals the preferential orientation of streptavidin in thin ice regions. Inset: typical 2D class averages showing predominantly top views. **f**, A representative cryo-EM image reveals streptavidin distribution in thick ice regions. Inset: typical 2D class averages showing predominantly top views and low-resolution side views. The results shown in panels (b), (c), (e), and (f) were independently reproduced in more than five micrographs with similar outcomes.



Extended Data Fig. 9 | Graphene reservoir enabling uniform liquid film in spray-based cryo-EM specimen preparation approaches. a-f, Schematics and typical cryo-EM grid atlases showing the sprayed droplets distribution on the traditional Quantifoil grid (**a-b**), Quantifoil grid with graphene (**c-d**), and graphene reservoir (**e-f**), respectively. Compared with traditional methods,

where sprayed droplets are often thick and uneven, the spontaneous liquid diffusion into graphene reservoirs results in much more uniform ice layers. Scale bars, 200 μm . All experiments were independently repeated on more than five grids, yielding similar results.

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Software and code

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Data collection	We used EPU (version 3.2.0) software, SerialEM (version 3.8) software and AutoEMation software to collect single-particle cryo-EM dataset. The AutoEMation software was written by Dr. Jianlin Lei at Tsinghua University. We used SerialEM (version 3.8) software equipped with PACETomo (version 1.4.4) scripts to collect tilt series.
Data analysis	We used MotionCor2 (version 1.1.0) to correct the beam-induced motion of cryo-EM micrographs and used CryoSPARC (version 4.6.0) and Relion (version 3.0.8) to perform 3D reconstruction. The CTF values of these motion-corrected micrographs were determined by CTFFIND4 algorithm (version 4.15) or PatchCTF in CryoSPARC (version 4.6.0). The structural analysis was performed in UCSF Chimera (version 1.16). The coordinates were generated in CryoNet.Refine (https://cryonet.ai/refine/). For cryo-ET reconstruction, we used Etomo (version 4.11) to align and reconstruct the tomograms. All these softwares are open-source.

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The cryo-EM density maps and coordinates of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin have been deposited into EMDDB under the accession numbers of EMD-66426, EMD-66423, and EMD-66424, and into the Protein Data Bank (PDB) under the accession numbers 26HH, 26DA, 26CW, respectively. The raw data of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin have been deposited into the EMPIAR, with the corresponding accession codes EMPIAR-13520, EMPIAR-13521, EMPIAR-13519, respectively. Any additional data supporting the findings in this manuscript are available from the corresponding authors upon reasonable request.

Research involving human participants, their data, or biological material

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Population characteristics

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Sample size

For graphene-sandwich encapsulation efficiency comparison, no less than five EM grids were characterized for each group, and thousands of squares were counted for each group.
For cryo-EM samples using ESI, no less than three EM grids were tested.
For cryo-EM analysis, the particle numbers used for the final reconstructions of LI.LtrB group II intron, SARS-CoV-2 spike protein and streptavidin were 179,409, 150,268 and 106,422 respectively, and the reported resolutions were 2.70 Å, 2.90 Å and 2.46 Å, respectively, estimated by the Fourier Shell Correction (FSC) = 0.143 cutoff criteria. In cryo-EM field, particle number used for final structural reconstruction varies greatly, which is well acceptable for such sample size used here.
To measure the particle motion, we carried out three particle-motion measurements of each samples, every of which was based on thousands of particles. Such example size and times of repeated experiments have been widely used to characterize beam-induced motion. For cryo-ET analysis to measure ice thickness, tens of tomograms were characterized for each condition.

Data exclusions

For cryo-EM reconstruction, particles grouped in bad classes with poorly defined features were excluded, because these particles were normally denatured or dissociated samples, which were harmful for high-resolution 3D reconstruction.

Replication

Particle-motion measurements were repeated three times for each group (as indicated in Methods), and to demonstrate the reproducibility of ice thickness control at different conditions, at least five grids were tested, all attempts at replication were successful with similar results.

Randomization

Samples were allocated random, including the particle-motion and structural determination. Representative tilt-series were taken in a randomized way, we stochastically sampled the areas to record.

Blinding

For cryo-EM reconstruction in Relion (version 3.0.8) and CryoSPARC (version 4.6.0), particles were randomly divided into two parts, and used for 3D structure determination. The consistence of structures generated by these two sub-datasets was used for the blinding test. The investigators were blinded to group allocation during data collection and/or analysis.

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